FinCalc Smart

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Scenario: US-MTG-20260616-8343

U.S. Mortgage Calculator

Personal Financial Scenario Report

United States · USD

fincalcsmart.com/us-mortgage-calculator

EXECUTIVE SUMMARY

Excellent Score

$360,000	$2,334.95	$480,583	$840,583
Mortgage Amount	Monthly Payment	Total Interest	Total Cost
	30-yr term	133% of principal	

PRINCIPAL VS. TOTAL INTEREST (OF TOTAL MORTGAGE COST)

Total projected: $840,583

Principal: $360,000 (42.8%)   Total Interest: $480,583 (57.2%)

AI-ASSISTED MORTGAGE SUMMARY

Your 30-year mortgage on a $450,000 property carries a loan of $360,000. At 6.75%, total interest over the loan term is $480,583 — 133% of the mortgage principal. Mortgage Health Score: 93/100 (Excellent).

Your Monthly (12×/year) payment of $2,334.95 covers principal and interest. Total estimated monthly housing costs are $2,934.95 (incl. property tax, home insurance).

Adding an extra $100/month toward principal could save $67,177 in interest and shorten your loan by approximately 3 years. A 200-basis-point rate increase to 8.75% would raise your monthly payment by $497.17.

INPUTS & ASSUMPTIONS

Home Price	**$450,000**
Down Payment	**$90,000 (20.0%)**
Annual Interest Rate	**6.75%**
Loan Term	**30 years**
Payment Frequency	**Monthly (12×/year)**
Annual Property Tax	**$5,400**
Annual Home Insurance	**$1,800**

MORTGAGE BREAKDOWN

Mortgage Amount	$360,000
Base Monthly P&I Payment	$2,334.95/month
Monthly Payment	$2,334.95
Total Monthly Housing Cost	$2,934.95/month
Total Interest Paid	$480,583
Total Mortgage Cost	$840,583
Balance After Year 5	$337,953
Balance at Midpoint (Year 15)	$263,864
Mortgage Health Score	93/100 (Excellent)

KEY DRIVERS / POSSIBLE ADJUSTMENTS

- Prepayment opportunity: adding $100/month toward principal saves an estimated $67,177 in interest and may shorten your loan by approximately 3 years.
- Rate sensitivity: a 200-basis-point increase to 8.75% would raise your monthly payment by approximately $497.17. Building a payment buffer or locking into a longer fixed term may reduce this exposure.
- Payment frequency: switching from monthly to accelerated bi-weekly payments effectively adds one extra monthly payment per year, reducing total interest and shortening your loan without a budget increase.

METHODOLOGY

What this calculator models:

- Uses standard US monthly compounding — divides the quoted annual rate by 12 to derive the monthly rate.
- Calculates the regular mortgage payment using the standard present-value annuity formula, then converts to bi-weekly or accelerated bi-weekly using the approved frequency multipliers.
- Applies PMI when down payment is below 20%, calculated monthly as (loan amount × annual rate) ÷ 12, and cancelled automatically when the outstanding balance reaches 78% LTV per the Homeowners Protection Act.
- Builds a year-by-year amortization schedule with extra monthly payments applied directly to principal.

What this calculator does not model:

- Closing costs, origination fees, discount points, title insurance, or other transaction costs.
- Adjustable-rate mortgages (ARMs) — this report models the full loan term at the quoted fixed rate.

EDUCATIONAL DISCLAIMER

This report is for illustrative and educational purposes only. Mortgage payment, interest, PMI, and Mortgage Health Score calculations are estimates and may differ from amounts quoted by a lender. US mortgage interest uses standard monthly compounding; actual payment amounts depend on your lender's specific calculation methods. PMI rates and cancellation timelines are estimates — actual removal is subject to lender verification of equity per the Homeowners Protection Act. DTI ratio guidelines (28% front-end, 36% back-end) are based on conventional Fannie Mae standards and may differ by loan program, lender, and credit profile. Results do not constitute financial, mortgage, lending, tax, or legal advice. Consult a licensed mortgage professional or financial advisor before making any mortgage or property purchase decisions.